Spatial organization of heterogeneities in the striatum

Marta Tataryn*[1], Salif Axel Komi[1], Kenji Morita[2], Rune W Berg[1], and Arvind Kumar†[3]

[1]Department of Neuroscience, Faculty of Health and Medical Sciences, University of Copenhagen, Blegdamsvej 3, 2200 Copenhagen, Denmark
[2]Physical and Health Education, Graduate School of Education and International Research Center for Neurointelligence (WPI-IRCN), The University of Tokyo, Tokyo 113-0033, Japan
[3]Division of Computational Science and Technology, School of Electrical Engineering and Computer Science, KTH Royal Institute of Technology, Sweden; Science for Life Laboratory, Solna, Sweden

October 28, 2026

Abstract

The striatum is usually described through a binary taxonomy of D1/D2 spiny projection neurons and coarse basal ganglia pathways supporting action selection and reinforcement learning. Yet the striatum is also a recurrent inhibitory network with strong neuromodulatory control, and its cellular and synaptic properties are not uniformly distributed. Spatial transcriptomics reveal continuous gradients in gene expression across striatal territory, superimposed on the discrete striosome–matrix division. Other studies report that intrinsic properties (such as excitability), inhibitory interneuron abundance and neuromodulatory fields (e.g. terminal distribution, clearance kinetics, receptor expression) vary in space. These spatially structured heterogeneities seem to align with striatal topographic input and output maps, suggesting that its computation cannot be uniformly distributed as well. We propose that the striatum is best understood as a spatial dynamical field, meaning that neighboring striatal microcircuits have similar but

*Corresponding author: marta.tataryn@sund.ku.dk
†Corresponding author: arvkumar@kth.se

not identical dynamical properties and these position-dependent parameters shape network dynamics affecting integration, competition, and plasticity differently in the striatal tissue.

Keywords: Striatum; neuron diversity; neuromodulation; interneurons; spatial gradients; recurrent inhibition; dynamical field.

Introduction

The basal ganglia (BG) consist of anatomically segregated subnuclei including the striatum, globus pallidus external (GPe), globus pallidus internal (GPi), ventral pallidum (VP), subthalamic nucleus (STN), substantia nigra pars compacta (SNc) and reticulata (SNr) (Figure 1, A). Virtually every domain of behavior from motor control to cognitive decision-making requires BG. From a top-down perspective, BG function is understood in terms of action-selection and motor control, both of which can be described in the framework of reinforcement learning. These generic functions are essential for any biological agent, and it is, therefore, not surprising that the BG constitute a phylogenetically ancient architecture (along with neuromodulatory systems) recognizable even in lamprey, one of the oldest living vertebrates (Robertson et al., 2014).

How exactly does the specific architecture of the BG give rise to such a wide repertoire of computations that span contexts, timescales and behavior, remains an open question. A canonical answer focuses on three pathways. At the coarse anatomical level the BG can be described as three major pathways: direct, indirect and hyper-direct pathways which can implement competition, gating, and stopping (Albin et al., 1989; Nambu et al., 2000, 2002; Gerfen, 2023). Interactions between direct and indirect pathways are modulated by the hyper-direct projections from cortex to STN, which together form the basic architecture needed for action selection (Gurney et al., 2001). In terms of afferents, striatum is the main input port of the BG, however STN and GPe also receive direct projections from cortical regions. These inputs determine the interactions among the three pathways to shape BG output and animal behavior (Gurney et al., 2001).

However, the pathway-centric picture largely renders the BG as a feedforward routing system. In particular, it ignores the recurrent connectivity within the striatum, and in the other nuclei. Indeed to understand the function, we need to know not only inputs and outputs, but also the local architecture of a system.

The striatum is an intricate inhibitory recurrent network composed of two dominant spiny projection neuron populations together with several classes of interneurons – all of which show substantial within population molecular and physiological heterogeneity (Cembrowski and Menon, 2018; Stanley et al., 2020; He et al., 2021). These neurons are subjected to multiple neuromodulators including dopamine (DA), acetylcholine (ACh), serotonin (5-HT), noradrenaline (NE), histamine, and endocannabinoids (eCBs).

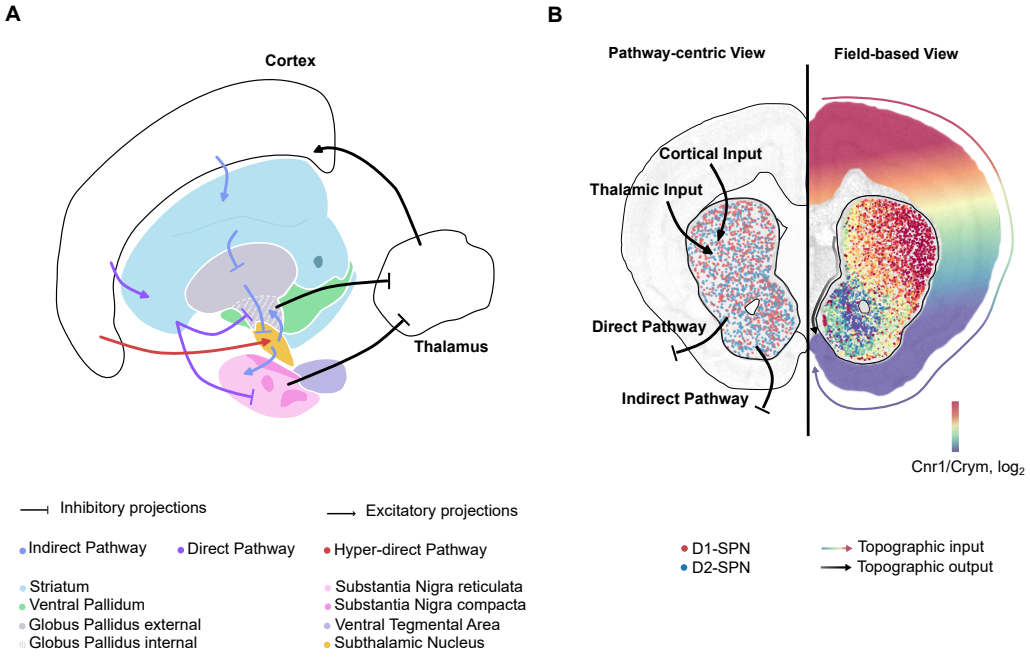


Figure 1: **From pathway-centric to spatially structured views of the striatum** **(A)** Canonical basal ganglia pathway diagram showing the direct, indirect and hyper-direct pathways. This schematic highlights only the classical pathways and it is not representative of the heterogeneous projection patterns, especially in the ventral striatum. Excitatory projections are shown with arrowheads and inhibitory projections with flat endings. **(B)** Comparison between a pathway-centric view and a spatial field view of the striatum. In the pathway-centric view SPNs are represented mainly by DA receptor expression and projection pathway. In the field-based view, SPNs are embedded in spatially structured continua of gene expression levels, topographic cortical input and topographic output. Cells are color-coded based on the Crym/Cnr1 gene-expression axis, computed from MERFISH-profiled striatal neurons (Yao et al., 2023). The field-based proposed perspective emphasizes that striatal computation may depend on position-dependent cellular and circuit properties.

Most cortico-thalamic and many subcortical regions project to the striatum (Wall et al., 2013) in a topographic manner. The specific convergence of sensory, motor, associative and motivational inputs in different striatal regions naturally suggests a multitude of integrative roles. Therefore, the striatum emerges as a major place for sensorimotor integration, combining cognitive state and reward contingencies to shape behavior (Reig and Silberberg, 2014). Several excellent reviews have discussed the anatomy and microcircuitry of the striatum in detail (Silberberg and Bolam, 2015; Burke et al., 2017; Gerfen, 2023). Nevertheless, functional implications of spatially structured heterogeneity in the striatal network have received relatively little attention.

The dominant striatal neurons are the spiny projection neurons (SPNs), also known as medium spiny neurons, and they differ not only by receptor profile and projection bias, but also in gene expression, intrinsic excitability, morphology, and likely local connectivity. Importantly, these features form spatial gradients aligned with structured input and output topographies. Furthermore, striosome and matrix regions may have a different distribution of neuronal diversity. In other words, the striatum is neither feedforward nor a homogeneous network of two types of inhibitory neurons.

This raises a basic computational question, largely absent from classic BG narratives: in a recurrent inhibitory network, does heterogeneity simply average out, or can it be exploited to produce position-dependent distinct dynamical regimes and learning rules?

To address this question, here we propose a bottom-up perspective in which striatal diversity and internal structure are not biological noise, but essential for its unique computational abilities. Specifically, we argue that the striatum is best understood as a heterogeneous, spatially organized computational substrate (Figure 1, B). Geometrically it is a manifold of microcircuits whose intrinsic properties, inhibitory connectivity structure, and neuromodulatory architecture jointly determine how sensorimotor and motivational information are integrated, selected, and learned. Striatum can also be abstracted as a spatial dynamical field, in which local population activity evolves over time under influence of spatially varying intrinsic properties, synaptic connectivity, and neuromodulatory states. In this framework, the spatial organization of cellular and circuit properties shapes the dynamics of neuronal activity. This new perspective of striatum is largely based on data from rodents (mouse). But we have also considered data from other species when possible.

Spatially Structured Heterogeneity

The cracks in the classic binary model of the striatum

GABAergic SPNs are the dominant (90–95%) cell population of the striatal network. Historically, SPNs have been divided into two classes based on their DA receptor expression (D1-like or D2-like) and projection preferences: direct pathway (dSPNs, mostly expressing D1-like DA receptors) and indirect pathway (iSPNs, mostly expressing D2-like DA receptors) (Gerfen, 2023). Hereafter, D1-SPN and D2-SPN refer to receptor-defined populations, whereas dSPN and iSPN refer specifically to direct- and indirect-pathway projection biases, respectively. Most D1-SPNs project directly to GPi and SNr (direct pathway) while sending collaterals to GPe (Cazorla et al., 2014) whereas D2-SPNs project to the GPe and then to the GPi/SNr (indirect pathway). In general, it is assumed that DA facilitates cortico-dSPNs connections and suppresses cortico-iSPNs connections (Shen et al., 2008). This view provides a basis for classical models of reinforcement-driven action-selection (Gurney et al., 2001; Frank, 2006).

However, chemical, morphological, biophysical, and projection details are challenging this simple two-population model of the striatum. With new single-cell and circuit tracing approaches it is now clear that a fraction of SPNs co-express both D1 and D2 receptors and create an additional indirect pathway which is strong enough to influence BG dynamics and function (Bonnavion et al., 2024). Even among dSPNs and iSPNs there is a high heterogeneity (see next section). Moreover, a tight correspondence between receptor type, pathway and function may be a good approximation in dorsal sensorimotor territories, but it breaks down ventrally (Fang and Creed, 2024).

The classic binary view treats the striatum as a feedforward network, ignoring its intrinsic connectivity. In reality, SPNs extend highly collateralized axons targeting SPNs within $\sim$100–200 μm, providing local recurrent inhibition (Planert et al., 2010; Taverna et al., 2008; Cinotti and Humphries, 2022). This organization has long motivated the view, based on anatomical work in rat, of the striatum as a lateral inhibitory network in which recurrent SPN collateral inhibition could contribute to competition between co-active populations (Groves, 1983; Wilson and Groves, 1980). Although individual SPN-SPN synapses may be weak, their cumulative recurrent inhibition and the type specific asymmetric connectivity are sufficient to create a decision-transition threshold to enable the classic action-selection dynamics, which can be tuned by DA and by the statistics of cortical inputs (Bahuguna et al., 2015).

SPNs Heterogeneity and the importance of space

The idea that striatal processing is spatially organized was already present in classical anatomical work, which described the neostriatum as a mosaic of repeated processing modules and multiple functional compartments defined by their input-output organization (Gerfen, 1992). Recent experimental data shows the diversity of SPNs in terms of gene expression, receptor composition, and intrinsic properties (Gokce et al., 2016; Stanley et al., 2020). Crucially, these features of SPNs are not randomly distributed across the striatum. Almost all described features of a SPN vary as function of spatial location within the striatum and neighboring SPNs have similar values of a feature. In other words, SPNs physiology is systematically different depending on where we look in the striatum.

Spatial transcriptomic studies first showed that small striatal domains share marker-gene signatures, forming molecularly defined territories (Märtin et al., 2019). Subsequent single-cell and spatial transcriptomic analyses further revealed that, superimposed in these territories, expression levels of many genes vary gradually across space giving rise to continuous molecular gradients in the striatum (Stanley et al., 2020; Roman et al., 2023; Yao et al., 2023).

To anchor these observations to a concrete spatial reference, a prominent expression gradient reported in SPNs is the Crym-Cnr1 molecular axis, which extends approximately along the dorsomedial-ventrolateral axis (Stanley et al., 2020; Roman et al., 2023). Other spatial transcriptomic datasets (Yao et al., 2023; Roman et al., 2023) have reported that many genes show some degree of spatial bias (Figure 2). Because a substantial fraction of these spatially patterned genes encode receptor subunits, ion channels and neuromodulatory components, these gradients are directly relevant to computation as they likely translate to spatial gradients in SPN excitability, plasticity and neuromodulator sensitivity.

Accumulating data suggests that biophysical gradients broadly align with molecular gradients to form a spatially structured parameter field across the entire striatum. For instance, transcriptomic patterns coincide with regional differences in electrophysiological properties (Roman et al., 2023). SPNs from different regions show systematic biases in multiple ion-channel gene families (potassium, sodium, calcium, leak, and HCN channels) (Roman et al., 2023). Direct physiological measurements also support the idea of spatial gradients. For example, intrinsic biophysical parameters of SPNs (membrane time constants, spike threshold, and input-output transfer functions) vary spatially, with the strongest contrasts between nucleus accumbens (NAc) shell and the caudate-putamen (CP) (Chuhma and Rayport, 2024). Similarly, SPNs in the NAc shell show higher input resistance and greater excitability than those in the dorsal striatum (Chuhma and Rayport, 2024; Lieberman et al., 2018). These spatial differences indicate that intrinsic

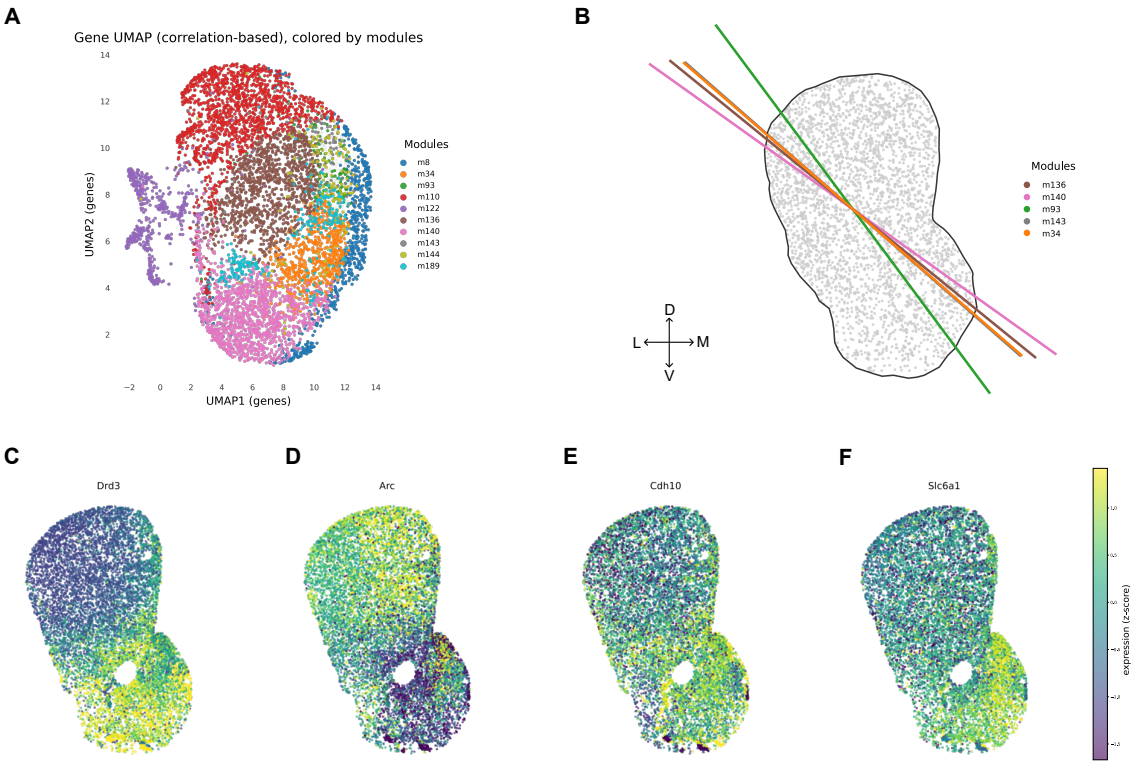


Figure 2: **Spatially patterned gene co-expression modules in striatal spatial transcriptomics atlas. (A)** UMAP embedding of genes based on similarity of their correlation profiles across striatal neurons in MERFISH-dataset (left hemisphere), (Yao et al., 2023). Points (genes) are colored by Louvain communities ("modules") detected in a thresholded positive gene–gene correlation graph. **(B)** Module spatial axes inferred from the relationship between module expression (PC1 of z-scored module genes) and CCF coordinates; spatial modules are sorted by global Moran's I. **(C–F)** Representative genes from example modules: **(C)** Module m136, **(D)** Module m140, **(E)** Module m93, **(F)** Module m134. For these modules, selected genes are shown as representative members of the module's co-expression program.

excitability is not uniform but varies smoothly across striatal territories.

Recent large-scale reconstructions reveal that morphology and connectivity of SPNs are also distributed in a spatially structured manner. For instance, a reconstruction of approximately 3000 SPNs revealed six morphological modules spanning contiguous striatal territories, each receiving distinct corticostriatal input patterns (Park et al., 2025). Complementary connectomics analyses have identified at least three connectivity-based SPN subtypes in the CP, implying spatially organized output motifs (Liu et al., 2025). This structured heterogeneity goes beyond the classic pathway-preference identity (D1/D2, direct/indirect) and supports a new perspective in which SPNs are characterized by multiple features (gene expressions, cellular properties) which are distributed along spatial gradients.

Taken together, these data support two related but distinct aspects of SPN spatial organization: 1. First, within the classical D1- and D2- defined populations, molecular, physiological, morphological and connectivity profile are position-dependent. 2. This variation is spatially correlated, such that neighboring SPNs tend to have more similar properties than spatially distant SPNs.

Interneuron diversity and nonuniform densities

Scattered among the SPNs are several classes of interneurons, including GABAergic interneurons and cholinergic interneurons (ChIs), which together constitute approximately the 5% of all striatal neurons. Like SPNs, interneurons also differ in their molecular identity and intrinsic physiology (Chuhma and Rayport, 2024). Importantly, their abundance and subtype composition vary across striatal territories. Despite their small population size, striatal interneurons can disproportionately affect local circuit activity through their extensive axonal arborization and strong connectivity. Spatial biases in their abundance or subtype composition can therefore generate significant regional differences in interneuron-mediated control across striatal regions (Lemos et al., 2026).

The classic catalog of striatal GABAergic interneurons includes: parvalbumin-positive (PV^+) fast-spiking interneurons (FSIs), SST/NPY/nNOS-expressing low-threshold spiking (PLTS / LTS) interneurons, calretinin-positive (CR^+) interneurons and a set of less abundant types such as tyrosine-hydroxylase-positive (TH^+) interneurons (Tepper et al., 2018; Burke et al., 2017). Single-cell transcriptomics has further expanded interneuron taxonomies in rodents (Muñoz-Manchado et al., 2018), and recent work in human dorsal striatum has identified up to eight interneuron classes and fourteen subclasses (Garma et al., 2024). Nevertheless, their diversity extends beyond taxonomy (i.e. cell types) to encompass spatial, biophysical, and synaptic dimensions. While several reviews have already discussed interneuronal identities in detail (Silberberg and Bolam, 2015; Burke

et al., 2017; Tepper et al., 2018, 2010; Tepper, 2010), here we focus on the evidence that inhibition due to interneurons varies systematically across space.

Quantitative mapping in mice reports that PV^+ FSI density increases systematically from ventral to dorsal striatum, reaching higher number dorsally (Bjerke et al., 2021; Van Zandt et al., 2024) (Figure 3, left). In contrast, their excitability decreases along the same axis (Chuhma and Rayport, 2024). This reciprocal relationship suggests that the dorsal striatum contains a denser but less excitable inhibitory lattice, whereas ventral regions contain fewer but more excitable FSIs. If we assume that the local connectivity is comparable, SPNs in the dorsal striatum would then share more feedforward inhibition than those in the ventral part. This suggests that inhibitory efficacy may be spatially tuned to the local circuit demands (Sreenivasan et al., 2022).

Recent work has extended this description within the mouse CP by providing a three-dimensional atlas of PV^+ interneurons, SST^+, and TH^+ in this region (Muhsinov et al., 2026). Importantly, the resulting spatial pattern is more complex than a gradient along one axis: along the antero-posterior axis PV^+ interneurons density increases towards the posterior edge, whereas TH^+ interneurons density declines posteriorly. SST^+ interneurons were likewise enriched in ventral CP, but followed a less monotonic pattern, with additional localized peaks in anterior, dorso-medial, lateral and medial tail regions (Muhsinov et al., 2026).

Outside the CP, however, the data about distribution of interneurons is fragmentary. For example, PLTS interneurons seem to be enriched in medial and ventral striatum, but quantitative stereological mapping remains incomplete (Van Zandt et al., 2024). Similarly, CR^+ and TH^+ interneuron distributions are biased in primate striatal compartments and along the rostro-caudal axis, however a complete quantitative map is still missing. Importantly, differences in cell-body density alone do not necessarily predict the strength of interneuron-mediated inhibition, which also depends on the spatial extent and density of their axonal projections and synaptic connectivity. Mapping these output patterns across striatal regions is essential for determining how spatial variation in interneuron populations translates into functional differences in local circuits. The available data already suggests that interneuron abundance, subtype composition, and connectivity may give different striatal territories distinct patterns of inhibitory control. These regional differences could alter the strength and timing of inhibition onto local SPNs and in turn shape how local populations respond to different input (Fino et al., 2018).

Neuromodulatory Architecture

Striatal networks are heavily controlled by neuromodulators. Neuromodulators regulate the gain of striatal neurons and circuits by changing the intrinsic excitability and synaptic efficacy. They also regulate synaptic plasticity, allowing the effect of previous activity

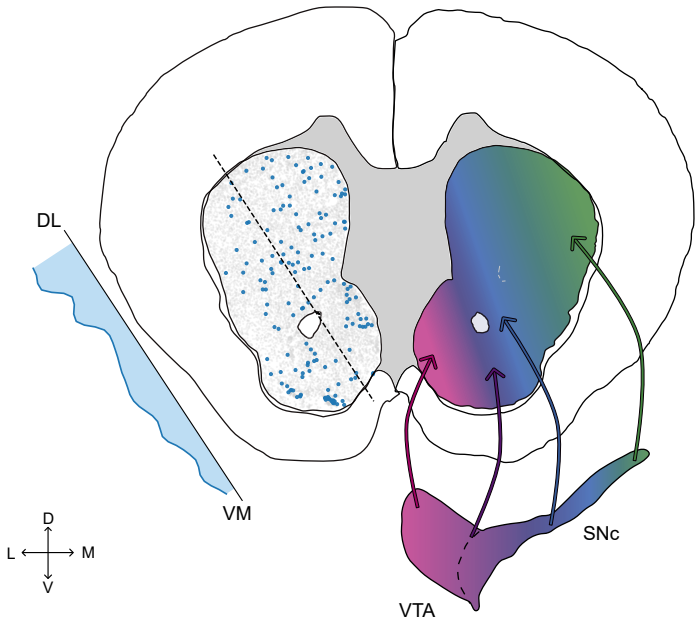


Figure 3: **Examples of inhibitory and neuromodulatory layouts. (Left)** PV^+ fraction profile along the dorsolateral–ventromedial (DL–VM) axis in a single MERFISH striatal section. PV^+ fraction (PV^+ /total) was computed from Gaussian-smoothed 2D binned cell counts and collapsed into DL–VM bins. Data is taken from Yao et al. (2023). **(Right)** Schematic illustration of spatially graded dopaminergic projections from the substantia nigra pars compacta (SNc) and ventral tegmental area (VTA) highlighting that dopaminergic projections in the striatum are topographically organized in the tissue, adapted from Chuhma et al. (2017).

and reinforcement to influence subsequent responses. Together these mechanisms can shift the operating regime of the local circuit without requiring changes in anatomical connectivity. The effect of neuromodulators depends on three main aspects: (1) where and how much they can be released (supply), (2) how quickly they are cleared or inactivated (extracellular kinetics), and (3) how strongly local cells and microcircuits respond, which depends on receptor distribution and localization, downstream signaling and circuit state. Data suggest that these three aspects of a neuromodulator action may also be distributed in a spatially structured manner.

In the striatum, DA and ACh are the two major modulatory systems that control circuit gain, excitability and plasticity but their effect is not uniform in the tissue. DA acts mostly through metabotropic receptors while ACh can act through both nicotinic and muscarinic receptors. Other neuromodulatory systems include 5-HT, NE, H and eCB and these also provide state-dependent modulation. Receptors for these modulators are expressed on both SPNs and interneurons (Gayden et al., 2023), with strong cell-type and spatial biases. In the following we mainly discuss the spatial gradients of DA and ACh.

Spatial gradients of dopamine

As a first approximation, substantia nigra pars compacta (SNc) and ventral tegmental area (VTA) projections preferentially innervate dorsal and ventral striatum respectively, but their projection fields overlap and include transitional zones across the SNc-VTA axis (Collins and Saunders, 2020). These long-range projections form the initial coarse spatial structure through their topography (Figure 3, right). Single-cell transcription profiling indicates that such coarse topography is implemented by multiple molecularly distinct DA neuron subtypes with biased projection fields (Poulin et al., 2020). Thus, different striatal regions integrate different mixtures of dopaminergic subnetworks, which can bias them toward different "reinforcement" statistics and can allow for the extraction of complex reward information (Wärnberg and Kumar, 2023; Millidge et al., 2024; Tsurumi et al., 2025).

Importantly, comparable activation of dopaminergic projections does not necessarily generate the same extracellular DA transient throughout the striatum (Calipari et al., 2012). Across species, evoked DA signals show a conserved dorsoventral gradient, which results in a larger release and faster uptake dorsally and smaller and longer-lived signals ventrally (Calipari et al., 2012). Consistent with this, in vivo measurements during self-paced exploratory behavior in an open field reveal faster DA dynamics in dorsal striatum and slower in ventral striatum (Jørgensen et al., 2023). This kinetic heterogeneity likely reflects regional differences in DA uptake, mediated by the DA transporter (DAT), release probability and the molecular identity of dopaminergic

afferents, and can be interpreted as a spatial map of "phasic bandwidth" (i.e., how faithfully brief DA transients are transmitted without being blunted by uptake and clearance) (Ejdrup et al., 2024). In addition, DA activity can also exhibit wave-like spatiotemporal patterns across dorsal striatum, indicating this signal is structured both by local kinetics and organized propagation dynamics (Hamid et al., 2021).

Other long-range neuromodulatory systems also show regional biases in their striatal axonal supply. Serotoninergic afferents from dorsal and median raphe broadly innervate the striatum, but with pronounced bias to the ventral part (Steinbusch, 1981; Di Matteo et al., 2008; Pommer et al., 2021). Noradrenergic terminals likewise densely innervate NAc shell, with sparse fibers in CP and NAc core(Berridge et al., 1997; López et al., 2024). Histaminergic fibers arising from the tuberomammillary nucleus innervate the striatum, but are relatively sparse compared to other forebrain targets, consistent with a role as diffuse state-dependent modulator (Bolam and Ellender, 2016; Lin et al., 2023).

Spatial gradients of ACh

ACh is special because it is mostly released locally within the striatum by striatal ChIs (Lim et al., 2014). Despite their low abundance, wide axonal arbors allow ChIs to regulate the striatal network state despite their small population (Ratna and Francis, 2025). A whole-striatum study shows that ChIs are more densely distributed in rostral striatum compared to caudal, and that observed local clustering is largely explained by underlying density gradients rather than random clustering (Carrasco et al., 2022). Patch-clamp investigations further reveal that their excitability decreases from ventral to dorsal striatum (Chuhma and Rayport, 2024). Finally, nicotinic ACh receptors (nAChRs) on DA axons control DA release, providing direct coupling between these two fields: cholinergic activity can trigger or amplify DA transients, shaping DA field (and waves) in both space and time (Matityahu et al., 2023).

Spatial gradients of neuromodulator receptor distribution

Receptor-subtype heterogeneity adds another layer of spatial modulation. A well documented example is the eCB receptor (CB1), which is highly expressed in the dorsolateral region of the striatum and the expression smoothly decreases along the dorsoventral axis (Van Waes et al., 2012; Stanley et al., 2020). Similarly, activation of the histamine H_3 receptor which reduces ChIs firing in the ventral, but not dorsal striatum (Varaschin et al., 2018). The dopamine D3 receptor also shows higher expression in ventral striatal domains relative to dorsal ones (Figure 2 C)(Gurevich, 2022). Serotonergic modulation seems to be also spatially structured at the receptor-expression level. For example, Htr1b transcription shows a dorsal bias in striatum (Yao et al., 2023). Together,

this experimental evidence establishes that neuromodulatory receptor expression and functional sensitivity are non-uniformly distributed spatially across striatal territories.

Interactions between different neuromodulators

Given the anatomy of the axonal projections and expression of receptors it is abundantly clear that neuromodulators overlap on the same neurons and therefore interact with each other. This is best described for the interaction between DA and ACh (Todd et al., 2025). ACh release can exhibit wave-like activity in the dorsal striatum (e.g., during salient events) (Matityahu et al., 2023). ChIs can also coordinate their network activity through DA-dependent polysynaptic interactions (Dorst et al., 2020). Conversely, ACh regulates DA release via nAChRs on DA axons, such that ACh dynamics can (and do) coordinate DA release across space (Matityahu et al., 2023), coupling ACh and DA signals spatially and perhaps leading to wave-like dynamics in DA release patterns (Hamid et al., 2021). However, it is not clear whether the interaction between DA and ACh (and among other neuromodulators) varies as a function of distance in the striatum.

In short, available experimental evidence suggests that neuromodulators influence computation through both spatial and temporal coordination signals, coupling distant striatal microcircuits. Systematic spatial mapping of many neuromodulators is still lacking, but the available evidence already shows that their projections and receptors have different spatial distributions. How these different patterns interact to shape striatal dynamics and plasticity remains an open question.

Striosome vs Matrix: Orthogonal Scaffold

Superimposed on the various continuous gradients are the striosomes interspersed within the so-called matrix (Brimblecombe and Cragg, 2017)(Figure 4, right). This patchy compartmentalization map is superimposed on, and largely independent of, the smooth dorso-ventral and medio-lateral gradients of expression we have described above, and it embeds both SPNs and interneurons in distinct input-output and modulatory contexts.

In primates, individual striosomes are patches on the order of a millimeter depending on the plane of section (Graybiel and Ragsdale Jr, 1978). Striosomes occupy the 10-15% of striatal volume in adult mammals and are characterized by high μ-opioid receptor expression and low calbindin immunoreactivity, whereas the surrounding matrix has the opposite pattern(Cirnaru et al., 2021; Graybiel and Matsushima, 2023). This neurochemical distinction is accompanied by the diverse expression of several other receptors and neuro-peptides, indicating that SPNs in striosomes and matrix experience distinct modulatory environments. Interneurons, including cholinergic and GABAergic

subtypes, are present in both compartments but show non-uniform distributions; for example, ChIs tend to be relatively enriched in the matrix and around striosomal borders (Bernacer et al., 2007; Ünal et al., 2011). Beyond marker differences, accumulating physiology indicates compartment-biased microcircuit function: SPNs in striosomes are intrinsically more excitable and have different sensitivity to DA compared with matrix SPNs (Prager and Plotkin, 2019; Prager et al., 2020; Miura et al., 2007). These compartmental differences likely reflect both cell-intrinsic properties and differences in dopaminergic innervation, receptor expression and local basal DA levels.

Projections from the cortex respect this compartmentalization to some extent. For example, limbic and associative cortices (e.g. prelimbic, orbitofrontal, insular, amygdala) preferentially innervate striosomes, whereas sensorimotor and much of the associative cortices preferentially innervate matrix (Crittenden and Graybiel, 2011; McGregor et al., 2019; Sadiq et al., 2025).

The striosome-matrix organization therefore adds a discrete compartmental structure to the broader continuous gradients across the striatum. These two forms of spatial organization can coexist; for example, a physiological property can vary gradually in space while also differing locally between striosome and matrix. Thus, when a broad spatial gradient intersects a striosome, the compartment identity may locally shift the value of that property relative to the surrounding matrix, without necessarily disrupting the larger-scale gradient. Whether this occurs, and for which molecular, physiological or circuit properties remains unclear. Direct evidence remains limited because relatively few studies have examined the same properties across both broad striatal gradients and the finer spatial scale of individual compartments. Functionally, they constitute neighboring microcircuits with different inputs, modulatory context, and outputs so the same cortical drive may be transformed differently depending on which compartment it enters. Moreover, striosomes can influence dopaminergic signaling, providing a control knob to shape dopamine signals affecting local excitability and plasticity. At the moment it is unclear how strongly striosome and matrix circuits couple to each other (directly and via interneurons), and whether compartment borders act as functional boundaries.

Input topography

The spatial heterogeneities described in the previous section are particularly relevant if the striatal inputs are themselves spatially structured so as to exploit them, and indeed they are. The striatum is a major subcortical convergence point where cortical, thalamic and limbic projections terminate in a highly convergent yet topographically organized way. As a result, each striatal domain is characterized by a specific combination of cortical, thalamic and subcortical projections it receives and the downstream targets

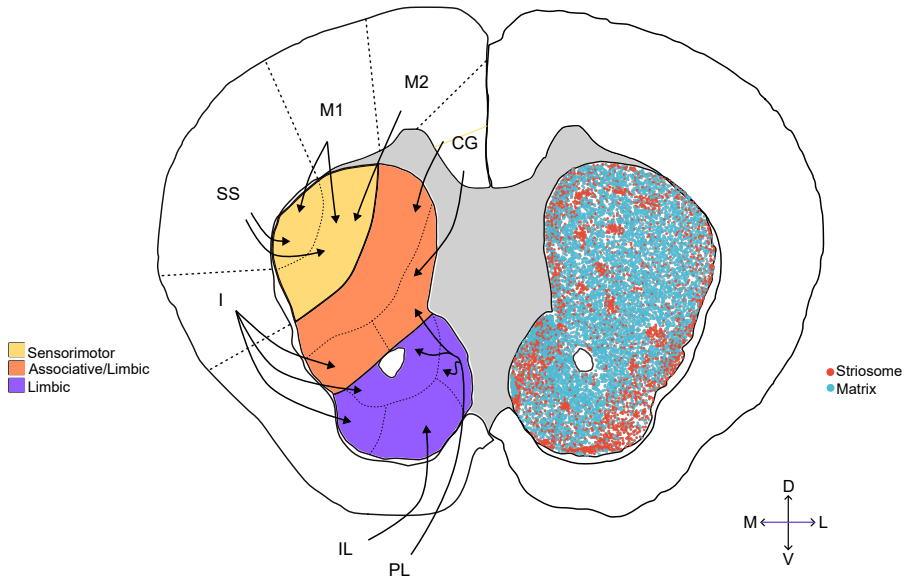


Figure 4: **Input-defined striatal domains and superimposed striosome–matrix scaffold. (Left)** Schematic coronal striatal section illustrating representative corticostriatal input territories used to define sensorimotor, associative and limbic striatal domains (SS, somatosensory cortex; M1, primary motor cortex; M2, secondary motor cortex; CG, cingulate cortex; I, insular cortex; PL, prelimbic cortex; IL, infralimbic cortex), adapted from Van Waes et al. (2012). **(Right)** Patchy striosome-matrix organization visualized by SPN spatial distribution colored by Sema5b expression (red; striosome-enriched); remaining SPNs are shown as matrix, data from (Yao et al., 2023).

it influences. Whole-brain anterograde mapping in mice shows that these connectivity patterns change gradually and continuously across the striatum (Hintiryan et al., 2016; Hunnicutt et al., 2016), providing a natural coordinate system that can align with the intrinsic and modulatory structure we just described (Figure 1, right).

Cortical regions project to the striatum in a continuous dorsolateral-ventromedial and rostro-caudal progression: sensorimotor cortex targets dorsolateral striatum; associative and prefrontal cortex target dorsomedial striatum, and orbitofrontal, insular, hippocampal and amygdala regions target the ventral striatum (Hunnicutt et al., 2016; Hintiryan et al., 2016; Haber, 2016). Projection fields from neighboring cortical areas overlap smoothly and heterogeneously. Quantitative studies show that this overlap is broader in the ventral striatum and sharper in the dorsolateral territory, which is consistent with a finer somatotopic organization in that region (Hooks et al., 2018). This topography still allows convergence of information from multiple sources into individual SPNs (Foster et al., 2021). This structured connectivity underlies the conventional assignment of dorsolateral striatum to sensorimotor/habit functions, dorsomedial striatum to associative/goal-directed functions, and ventral striatum to limbic-motivational functions, while still allowing substantial overlap and interaction between these domains (Figure 4, left).

Along the cortex-defined connectivity axis the mixture of thalamic contribution also changes systematically. Intralaminar and related nuclei (parafascicular and centrolateral in rodents) project preferentially to dorsolateral striatum with a strong matrix bias, whereas midline and mediodorsal nuclei project more strongly to medial and ventral striatum (Smith et al., 2009, 2014; Martel and Galvan, 2022). These thalamostriatal projections are themselves topographic and mostly matrix-directed, and they tend to target the same striatal territories that receive the matching cortical inputs (Huerta-Ocampo et al., 2014). As a result, any given point in the striatum receives a position-dependent mixture of cortical and thalamic signals, and neighboring points receive partially overlapping mixtures. This defines a connectivity gradient that is aligned with, and can in principle exploit, the intrinsic and modulatory gradients described above.

Output topography

Striatal activity is routed through partially segregated output channels that largely preserve the input topography, and it differs between matrix and striosome compartments and between dorsal and ventral domains (Foster et al., 2021; Gerfen, 2023).

In the dorsal striatum, matrix SPNs form the canonical direct and indirect pathways (dSPNs to GPi/SNr and iSPNs to GPe) preserving the topographical order of the cortical and thalamic inputs and relaying back to the matching cortical fields via thalamus (Gerfen, 2023; Hintiryan et al., 2016; Hunnicutt et al., 2016). By contrast, in the ventral striatum, both D1- and D2- expressing SPNs innervate the ventral pallidum (VP). The VP in turn projects to mediodorsal and related thalamic nuclei, as well as to lateral hypothalamus, lateral habenula, and midbrain structures, including VTA and SNc (Foster et al., 2021). Across these pathways, topography is broadly maintained because neighboring ventral striatal territories tend to project to neighboring VP and thalamic domains, and on to partially corresponding cortical territories (Foster et al., 2021).

Striosomal SPNs have privileged access to dopaminergic neurons in SNc and VTA, and in rodents they also project to specific GPe subdomains, anatomically distinct from the matrix motor outputs (Crittenden and Graybiel, 2011; Graybiel and Matsushima, 2023; Lazaridis et al., 2024). Importantly, matrix projections to SNc have also been reported, but striosomal ones seem to be particularly specialized (Smith et al., 2016; Lazaridis et al., 2024). Striosomal projections are themselves topographically organized along the dorsoventral axis: striosomes in the ventral striatum more strongly innervate the VTA and those in the dorsal part are biased towards the SNc. Thus, we can infer that the same gradient that organizes cortical input is mirrored in the mapping from striosomes to dopaminergic midbrain populations.

Functional Implications

Functional implications of the spatial distribution of neuron properties

The gradients in molecular and biophysical properties of SPNs suggest that each spatial voxel of the striatum implements a distinct transfer function and spike pattern. Regions with higher input resistance and smaller leak conductances (i.e., longer membrane time constant; prominent in ventral striatum) would be more permissive: they can respond to weakly correlated inputs and are, therefore, better suited for temporal integration. Conversely, regions where SPNs have smaller time constants (e.g. dorsal striatum) would be more selective, requiring stronger and/or more correlated inputs to reach spike threshold. Similarly, other SPNs biophysical properties may render different striatal regions selective to specific input statistics. In other words, even similar input patterns may be transformed differently across striatal territories because their intrinsic properties differ.

Accumulating evidence indicates that each brain region has different spike statistics (Shinomoto et al., 2009; Mochizuki et al., 2016): e.g. neurons in the early sensory areas show highly irregular spiking activity, those in the motor regions more regular patterns, and neurons in the associative areas more bursting activity. It is reasonable to hypothesize that the spatial organization of cortical and thalamic inputs may align their different temporal input statistics with the different intrinsic integration properties of their striatal targets to preserve information flow and maximize discriminability. For example, SPNs with longer membrane time constants can integrate inputs over larger temporal windows, whereas SPNs with shorter time constants integrate over shorter windows and are in turn more sensitive to temporally close input. Since the striatum integrates information from many cortical and thalamic regions, intrinsic heterogeneity may provide a palette of integration modes that could decode diverse input statistics.

However, this hypothesis has a critical dependency: to exploit neuron diversity gradients, both input projections and recurrent connectivity should be spatially structured such that each striatal region is attuned to the input statistics of different cortical regions. If cortico- and thalamo–striatal projections do not respect these intrinsic gradients, local diversity would have limited functional consequences. By contrast, if projections are biased such that cortical areas with specific spike statistics preferentially target striatal territories with matching neuron properties, then the striatal heterogeneity of SPNs becomes a crucial design feature. Given the gradients of neuron biophysical properties, we argue that even before lateral inhibition comes into play, some striatal regions are intrinsically biased to “win” or “lose” under the same input statistics because their intrinsic thresholds differ. Graded intrinsic properties pre-configure where cortical and

thalamic information can win in the striatum. In practical terms, territories containing intrinsically more excitable SPNs may be recruited more readily than territories containing less excitable SPNs under comparable drive. Thus, spatial variation in intrinsic properties can create position-dependent recruitment thresholds even before recurrent inhibition comes into play.

Functional implications of fast-spiking interneuron diversity

Interneurons in the striatum compensate for the small population size (1% or less for each type) by forming strong synapses with high connection probability. Even if interneurons were randomly distributed with a uniform mean density, their small numbers would result in local fluctuations in cell density. Such a spatial distribution could form local "hot-spots" where their effect on SPNs is disproportionately large.

The high connectivity of FSI in the striatum results in a feedforward inhibition (FFI) which is qualitatively different from the cortical FFI: the only 1% FSIs in the striatum implies massive sharing of FFI in the striatum (Burke et al., 2017) as compared to the cortex. Computational models suggest that given high FSI→SPN connectivity, FSIs can promote local synchrony among SPNs (Yim et al., 2011; Guo and Kumar, 2026). Because a relatively small number of FSIs project to many SPNs, their activity generates shared inhibitory inputs across the SPN population. Mathematically, this introduces correlations in the FFIs and therefore, introduces a 'low-rank' component of inhibition, since the dimensionality of FSI-mediated input is limited by the number of contributing FSIs. Such shared inhibition could coordinate SPN activity across local clusters.

Given the opposing spatial gradients in FSIs density (higher dorsally) and excitability (higher ventrally), one might expect some compensation in the average inhibitory drive onto SPNs. However, even if the average inhibition were similar across regions, assuming comparable connection probabilities, SPNs in dorsal regions would receive more shared feedforward inhibition than those in ventral striatum. This reciprocal gradient of FSI density means that dorsal regions possess a denser but less excitable inhibitory lattice, resulting in a strong suppression that stabilizes activity patterns, whereas ventral regions have sparser but more excitable FSIs. Therefore, spatial gradients of FSIs would manifest as gradients of not only excitability but also synchrony and trial-to-trial variability (Guo and Kumar, 2026).

Furthermore, dorsal territories, with dense FSI networks, may favor rapid, decisive selection and precise timing and ventral regions, with sparser and slower inhibition, may allow longer integration and exploratory combinations of inputs. At intermediate zones and compartment borders, anisotropic inhibition can bias how activation spreads, guiding sequences of activity through the striatum.

How the FSI effects on neuronal activity dynamics discussed above may manifest at

the behavioral level is unclear but some interesting suggestions are emerging. Kato and Morita (2025) have recently shown that assuming a gradient of the rate of forgetting (and learning) could explain the different DA release patterns (lateral abrupt vs medial ramping) observed experimentally. They suggested that the regional difference in the forgetting/learning rate could fit distinct computations of hierarchical or distributional reinforcement learning algorithms. Spatial gradient of PV+ FSI (and perineuronal nets) could form a possible neural mechanism that controls the gradient of forgetting in the striatum by regulating the degree of plasticity. Other high-level functional models of the striatum should be further examined to understand how spatial gradients in the neuromodulators and neuronal properties could shape function.

Beyond FSIs, the PLTS class provides a complementary mode of inhibition over longer windows, because they are characterized by high input resistance, and prolonged depolarizing plateaus and they preferentially target distal dendrites of SPNs (Tepper et al., 2010; Tepper, 2010). As a result, their influence is slower and more modulatory than that of FSIs because they influence subthreshold integration and gain rather than precise spike timing. Whether PLTS interneuron density or intrinsic excitability systematically vary across striatal sub-regions remains to be established. If indeed PLTS interneurons were more prevalent or more excitable in ventral/medial than dorsal/lateral territories, one would predict longer integration windows in those zones, but this remains a hypothesis.

New dynamics and function may arise due to spatial gradients

The heterogeneities we have described above remain either fixed or change slowly over time. Such heterogeneity is called "quenched noise". Such noise is known to reduce synchrony and make the asynchronous-irregular activity state more robust (Kumar et al., 2008; Hutt et al., 2023; Dahmen et al., 2025). However, this view assumes that neuron properties are uniformly distributed in the network in an unstructured manner, such that in every network subregion we will find similarly diverse types of cells.

If we assume that heterogeneities are spatially distributed according to some rule, novel dynamical properties and activity patterns may emerge in the network. This is best illustrated by a network model in which inhibitory neurons form connections in a distance-dependent manner but every neuron has a slight preference to make extra connections in a given direction. When neighboring neurons have similar direction preferences (but far away neurons have unrelated direction preferences), sequential activity emerges in a network which would generate stable activity bumps (Spreizer et al., 2019). From a mechanistic perspective, this provides a way of generating ordered trajectories of activity in the striatal territory, supporting, for example, learned action sequences, state transitions without an external pre-configured driver (Markowitz et al.,

2018; Cox and Witten, 2019; Sheng et al., 2019).

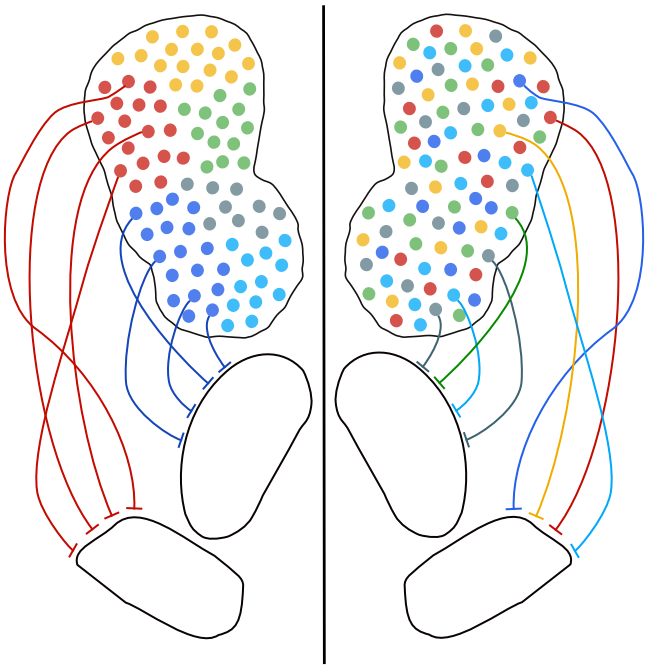


Figure 5: **Why spatially structured heterogeneity matters. (Left)** Spatially correlated heterogeneity (similar clusters) produces aligned, region-specific input–output mapping: nearby units implement similar transfer functions and project in a coherent, topography-preserving way. **(Right)** If the same heterogeneity is spatially shuffled, projections are effectively random across space and their effects cancel at the population level, approximating a homogeneous network.

The insights from Spreizer et al. (2019) can be extended from connectivity anisotropy to the spatial distribution of neuron properties. That is, the different biophysical and chemical gradients we have discussed above could structure the network to produce non-trivial and novel activity patterns.

Furthermore, gradients in neuron and synapse properties will have an effect on downstream readouts (i.e. globus pallidus). For instance, in a network with spatial connectivity, if the neighboring neurons have different properties their effect will be averaged out due to recurrent interactions. Moreover, if the downstream network is also reading activity in a spatial manner, downstream neurons will sum up the diverse spike patterns and individual neuron spike patterns will be averaged. This assumption is actually supported by anatomical studies in macaques showing that striatopallidal projections are highly topographic, meaning that neighboring striatal territories project to corresponding pallidal territories, while projections from distinct striatal regions remain largely non-overlapping (Heilbronner et al., 2018). At the same time, anatomical studies across species also indicate a distinctive pallidal organization in which pallidal neurons receive convergent striatal input along their extended dendrites (Haber and Nauta, 1983; DiFiglia and Rafols, 1988). Together, these anatomical features suggest

that pallidal readout is not only spatially structured but also convergent, preserving regional differences in striatal activity while integrating signals within those territories. In this setting, if neurons with similar (not identical) properties are spatially clustered, recurrent interactions may further amplify the neuron dynamics and the network will have regions with significantly different spiking patterns (Figure 5). Therefore, to exploit the computational advantages of neuron diversity it is important that neuron properties are distributed in a spatially correlated manner.

Spatial gradients of neuromodulation

Because DA, ACh and eCBs all influence corticostriatal and intrastriatal plasticity, their spatial organization implies that also plasticity rules may vary in the striatal tissue. Dorsal striatum, with higher evoked DA release and faster uptake, likely operates with higher gain and shorter integration times. This places dorsal territories closer to selection thresholds, producing rapid, high-contrast choice under cortical drive. Ventral striatum, with longer DA transients may favor accumulation and valuation over longer windows. This interpretation should be treated as a circuit-level hypothesis because dopaminergic axons may also influence local activity through co-transmission and through local interactions with glutamatergic, GABAergic and cholinergic mechanisms (Cachope and Cheer, 2014; Chuhma et al., 2018; Roberts et al., 2021; Matityahu et al., 2023). The different spatial distribution of ChIs implies that rostral and caudal striatal territories more readily broadcast ACh transients that recruit nAChR-dependent DA release, effectively lengthening integration windows and facilitating "credit assignment" for temporally extended cues. Wave-like DA and ACh dynamics transiently synchronize distant striatal domains, aligning their dynamics so that multiple "action channels" switch together during salient events or state transitions. Functionally, this is a temporary reduction in dimensionality that enables coordinated reconfiguration. Interestingly, because eCB receptors influence short- and long-term plasticity at both glutamatergic and GABAergic synapses, their spatial distribution suggests that plasticity rules also may differ systematically within the striatum.

Taken together, intrinsic gradients, inhibitory microcircuits, and neuromodulatory kinetics lead to position-dependent operating regimes for striatal computation. Functionally, this suggests that the striatum does not implement a single canonical "action-selection" computation everywhere, but rather a family of related computations whose operating regime depends on position. Dorsal sensorimotor territories may be biased toward fast, high-contrast selection with tight temporal control, supporting rapid commitment and stable execution, whereas ventral and medial territories may be biased toward longer integration windows and greater sensitivity to contextual and motivational variables, supporting valuation and flexible updating. Intermediate zones may act as

transition regions where competition, gating, and sequencing interact, enabling graded arbitration between habitual and goal-directed control and still preserve topography. In this view, spatial heterogeneity provides a principled substrate for parallelism: multiple corticostriatal loops can now run similar algorithms under different dynamical constraints.

Outstanding questions

Several questions remain open. How are spatial heterogeneities established and shaped by development, learning and disease? How do different gradients interact with each other and with recurrent connectivity? Are cortical and thalamic input statistics aligned with the local integration and plasticity features of their striatal targets? We elaborate on these questions below.

One main implication of this proposed perspective is that heterogeneity is not just a source of robustness to perturbations or noise reduction but a structured computational resource that can be exploited by the network. In particular, differences in the spatial distribution of intrinsic excitability, inhibitory control, and neuromodulation can create different striatal territories that differ in how they integrate inputs, compete and undergo plasticity. Thus, heterogeneity can effectively generate distinct local operating regimes. Therefore, it is important to understand how development, learning and disease processes shape the heterogeneity landscape of the striatum.

Next, it is important to understand how different gradients interact with each other and whether they amplify or cancel one another. It is possible that some gradients may push circuit output in opposite directions, for example, higher excitability in one territory may be offset by stronger inhibition or different neuromodulatory gain. Thus, simple summary measures (e.g. mean SPN firing rate or average inhibitory tone) may appear similar across regions. Yet, similarity in average population activity does not imply similarity in computation. In other words, intrinsic properties, inhibitory circuits, and neuromodulators shape different aspects of network behavior, including trial-to-trial variability, shared correlations, the timing of competition, the effective integration window, and the induction of plasticity.

Another open question is whether the spatial gradients of neuron properties are similar in D1- and D2 SPNs. For example, do gradients in intrinsic excitability and gene expression follow the same spatial axes in both populations, or are they organized differently?

The functional relevance of these spatial gradients is also contingent on the spatial structure of recurrent connectivity among the SPNs and other interneurons. At the moment very little is known about spatial structure of connectivity. We need to estimate

local connectivity in different regions of the striatum without assuming that all matrix and striosome compartments follow similar connectivity rules.

In the absence of these data we are forced to assume that local wiring motifs such as distance-dependent SPN collateral inhibition and the divergence/convergence of interneuron outputs—are broadly conserved across striatal regions, so that density or intrinsic excitability gradients translate into predictable changes in effective circuit dynamics. With such connectivity, the spatial gradient will be largely preserved and affect the function in a rather predictable manner as discussed above.

However, if the connectivity parameters vary systematically across regions, they could counterbalance density or excitability gradients and alter the resulting functional picture. Likewise, whether cortical and thalamic inputs are aligned with local integration regimes—for instance, whether the temporal statistics of input activity match the intrinsic time constants of their striatal targets—remains largely unknown.

Importantly, this framework has implications for how we think about learning, flexibility, and disease. If neuromodulatory kinetics and receptor landscapes define spatially varying learning rates and eligibility windows, then plasticity rules themselves may differ across the striatum, biasing what is learned and how quickly in different territories. Therefore, it is important to measure the spectrum of learning rules and synaptic malleability landscape of the striatum.

Differences in plasticity could help explain why habits, goal-directed actions, and motivational valuation show distinct learning dynamics and differential sensitivity to neuromodulatory dysfunction. Moreover, viewing the striatum as a structured field suggests that neurological and psychiatric disorders may preferentially disrupt specific spatial regimes rather than uniformly altering striatal function.

On the computational modelling side, a new generation of striatum models are needed to understand in what way the landscape of neuron/synapse properties may be reflected in the neuronal activity dynamics. Such models will not only synthesize experimental data but they can simplify tedious experiments by suggesting new measures of network heterogeneity and activity.

Conclusions

Spatial gradients in neuron properties give rise to specific striatal regions with different integration and selection regimes conferring them position dependent time constants, thresholds and nonlinearities which results in a spatial bias in how cortical and thalamic inputs are transformed locally. Superimposed on this, spatially structured inhibition in the form of non-uniform densities, excitability and connectivity, shapes competition, shared variability and temporal precision in a location specific manner. Finally the

firmware of striatal networks is modulated in both space and time by smoothly varying neuromodulatory fields. Alongside these continuous gradients, striosome-matrix compartmentalization introduces a discrete spatial organization that may further differentiates local input, modulatory and output contexts.

In conclusion, we propose that incorporating spatial localization explicitly into models of the striatum is not just a matter of anatomical detail but a crucial conceptual shift. Indeed, the striatum should be modeled and interpreted as a spatially organized substrate in which these spatially structured heterogeneities jointly shape computation and are crucial for striatal function. Embracing such a view offers a path toward unifying multimodal observations and toward understanding how a single anatomical structure can support such a wide repertoire of functions across contexts and timescales.

Conflict of Interest Statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Funding

MT was supported by Neuroscience Academy Denmark under Lundbeck Foundation grant R389-2021-1596. AK was supported by StratNeuro SRA grant from Swedish Research Council.

Acknowledgments

We thank Gilad Silberberg for helpful discussions during the preparation of this review and Ainoa Konomi Pilkati for helpful comments and feedback on the manuscript.

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